

Things You Wanted to Know!

Q1. What does mAh mean? I would like to know how many hours will Mircomax A120, which features a battery of 2000mAh, work for before it drains out? How can I calculate hours of working on the basis of mAh?

Ismail Taibani
on email

A1. The capacity of a battery is rated as amp-hour current (expressed as Ah). This is the amount of power, in amps, a battery can deliver over a specific period of time. If a battery has a rating of five amp-hours, it can provide up to 5A continuously for one hour, or 1A for five hours.

The current rating of a battery is as important as its voltage rating. It specifies the maximum current that could be drawn at the rated voltage. Ah is a large unit, therefore for capacity of a smaller battery, mAh is used. mAh means milliamp-hours. As milliamp is 1/1000th of an amp, 1000mAh can deliver 1000 milliamps for one hour, or 1-amp-hour. 2000mAh means it can deliver 2 amps for an hour.

The drain current of the battery depends on mobile as the load. If the mobile consumes maximum load current of 200mA, it will work up to 2000mAh/200mA = 10 hours.

Q2. Why do some mobile phone batteries have four terminals and others have only three? What are the functions of third and fourth terminal pins?

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A2. Nearly all mobile phone batteries have three terminals and sometimes four terminals.

Battery pin descriptions are given below:

GND or negative terminal. A reference point for measuring the other voltage or returning point for electric current. It is the same as a negative battery terminal.

BAT+ or positive terminal. A positive terminal is also known as a positive battery terminal.

Both positive and negative terminals are duly-marked on a battery.

Battery temperature (BTemp). This terminal is used for sensing. Batteries are charged by applying controlled current through the charging circuit. While charging a battery, some chemical and electric energy is converted into heat, so a battery tends to overheat while getting charged. Overheating and overcharging may damage the battery permanently, shorten its life or lead to explosion. Thus, we need to control the heating of the battery during charging.

BTemp terminal of the battery is used to sense the temperature of the battery, and accordingly, the charging circuit of the mobile phone changes the voltage levels of the output. This change in voltage by the charger circuit controls the charging current to the battery. The battery-charging circuit continuously senses this terminal, thus temperature of the battery while charging, and accordingly, varies the current or the charging rate.

Battery site indicator or battery system indicator (BSI). There are some mobile phones that have shorter BSI pins. This is because when a battery is removed and the phone is switched on, the BSI circuit interrupts first, which tells the processor to shut down all communication to prevent damage.

The battery block contains BSI resistors for battery identification. BSI fixed resistor value indicates the chemistry and default capacity of a battery. Temperature and capacity information is needed for charge control (charging current and voltage). It is connected to the resistor to GND. The phone measures the resistance of BSI-GND connectors and determines what kind of battery is in the phone.

For example, Nokia NiMH batteries have BMC-2 640mAh 3.3kΩ, BMC-3 900mAh 5.6kΩ.

Q3. I have two questions. First, what is the difference between ARM processor and ARM Cortex? What is Arduino Uno R3 processor?

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A3. **ARM and ARM Cortex.** These are both processor cores on which microcontrollers (MCUs) are made. ARM Cortex-M3 is an implementation of ARM7v architecture. A number of manufacturers are producing MCUs based on Cortex-M3 and ARM7 cores. ARM Holdings licenses the design to manufacturers that produce their own implementation.

While ARM7 implements both ARM and Thumb instruction sets, Cortex-M3 supports only Thumb-2 instruction set. Cortex-M3 is aimed at the deeply embedded market. It is designed to be low-cost and low-power and provides good performance for the cost and power, particularly suited to automotive and wireless communication applications.

The other ARM core is ARM7TDMI (and ARM7TDMIS). MCUs based on this core are, in many ways, the workhorse of the embedded world.

Arduino Uno R3 processor. This is a MCU board based on ATmega328. It has 14 digital input/output pins (of which six can be used as pulse-width modulation outputs), six analogue inputs, 16MHz ceramic resonator, USB connection, power jack, ICSP header and a reset button. It contains everything that is needed to support the MCU; simply connect it to a computer with a USB cable, or power it with an AC-to-DC adaptor or battery, to get started.

Answers compiled by EFY joint director (training), Col. N.C. Rinde (Retd.). Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender